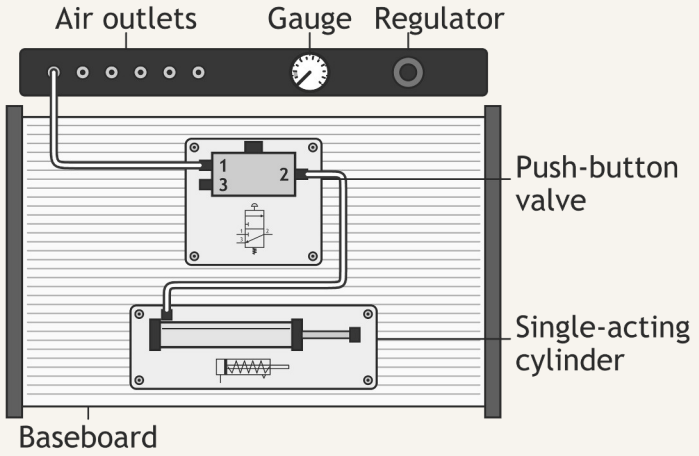
# Daily Review

| | | |
|---|---|---|
| 1 | Name one thing air can move. | Bus doors, a tyre pump |
| 2 | Three parts of a system? | Input, process, output |
| 3 | Which engineer works on moving parts? | Mechanical |
| 4 | Energy of movement is called? | Kinetic energy |
| 5 | Energy is measured in? | Joules (J) |

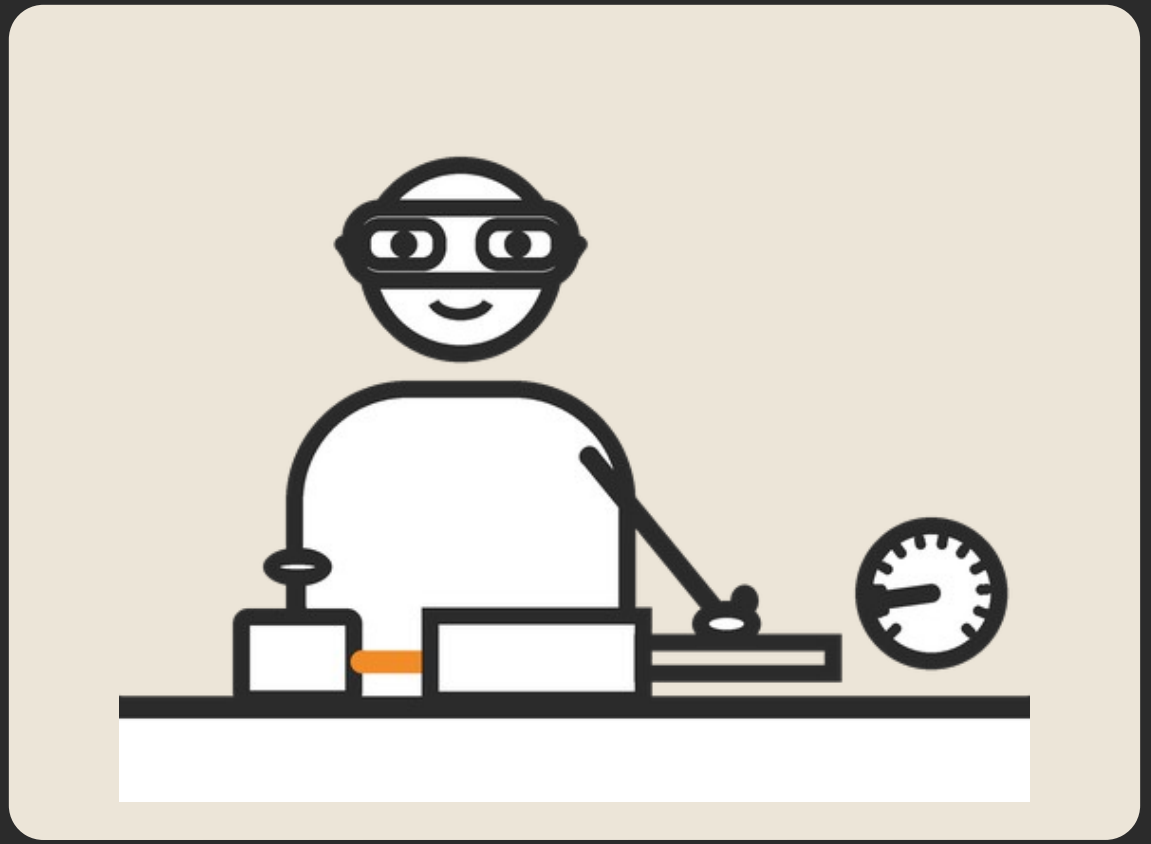
# Safety First and Last

How do you make a park gate move with air, safely?




S3 Engineering Science

# Squashed air can push hard. How do we use it safely?



The danger: a hand on the rod

PART 1

# Safety First and Last

Learn to use air safely, then build your first air circuit.

# By the end of this part…



1. I can follow the safety rules and the regulator routine
2. I can say what the rod does when I press and let go
3. I can test a circuit in the simulation before I build it

SECTION A

# Safety First

Air can push with a big force. What keeps us safe?

# Air Can Push Hard

## Pneumatics

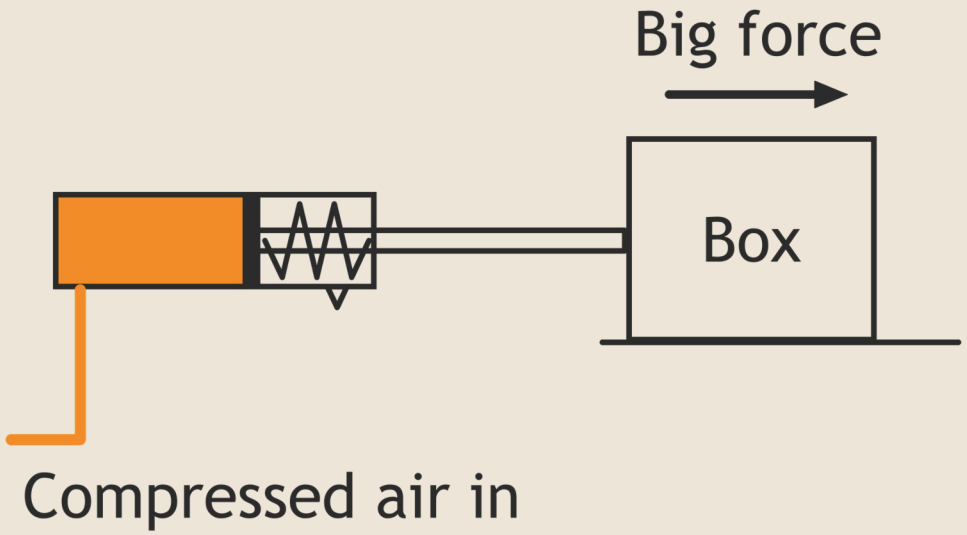
using compressed air to make things move

## Compressed air

air squashed into a small space

## Why take care

it pushes with a big force. It can hurt eyes, ears and skin.



# Where You Meet It: Air Tools

## Air tools

garages undo wheel nuts with air

## Why air

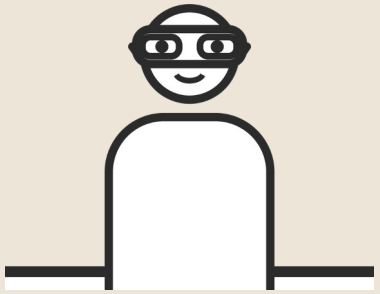
it pushes hard and fast

## Take care

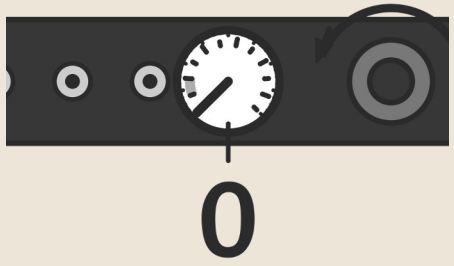
the same five safety rules



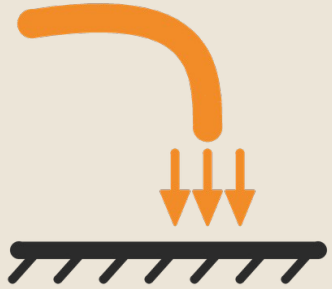
# The 5 Safety Rules (1 of 2)



**1 Goggles on when you build and when you test.** A tube can blow off near your eyes.




**2 Regulator at 0 while you build or change a circuit.** Parts can move while your hands are in.

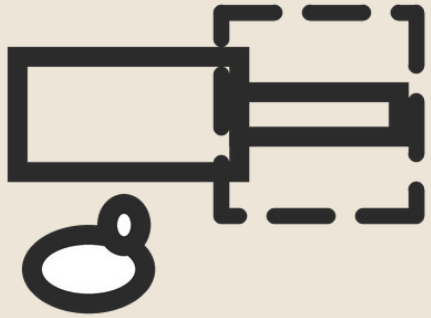


**3 Never point air at anyone.** Tube ends point down at the bench.

# The 5 Safety Rules (2 of 2)



**4 Hear a hiss? Regulator to 0 at once.** A hiss is a leak. The tube may blow off.



**5 Fingers clear of the piston rod.** It moves out with force. It can trap fingers.

# The Gauge: Between 1 and 2 Bar

## Read

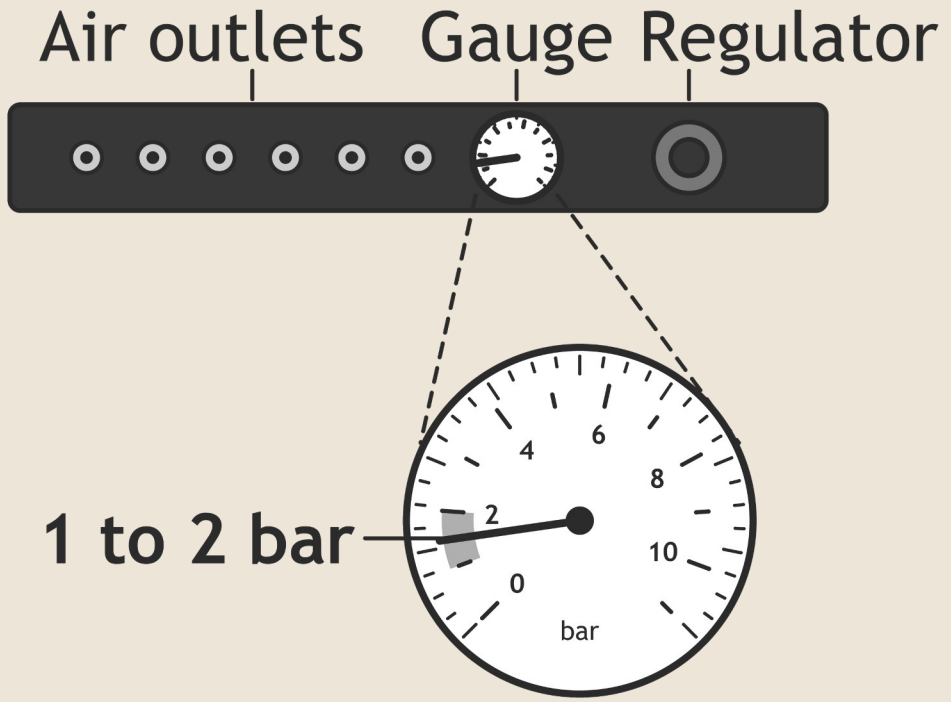
the inner numbers. They are in bar.

## Set

between 1 and 2 bar. Never above 2. Never below 1.

## Change anything?

turn it back to 0 first.



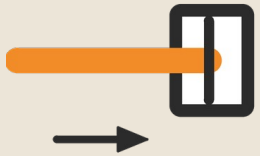
# The Regulator Routine



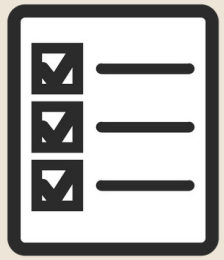
**1 Goggles on**



**2 Regulator at 0**



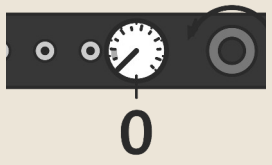
**3 Build**
push every tube in fully



**4 Air Permit signed**



**5 Set 1 to 2 bar**
then test




**6 Back to 0**
regulator to 0 when you finish

# Show Me: Safety Quick-Fire

| | | |
|---|---|---|
| 1 | Before you change a tube, the regulator is at… | 0 |
| 2 | You hear a hiss. What do you do first? | Regulator to 0 |
| 3 | Where do tube ends point? | Down at the bench |
| 4 | What pressure do you set for testing? | Between 1 and 2 bar |
| 5 | When do you wear goggles? | When you build and when you test |

You want to move a valve. The air is on. What do you do first?



A move it quickly

B turn the regulator to 0

C ask a friend to hold the cylinder

D press the button to empty it

Answer B: Regulator to 0 first, every time you change anything.

# Safety Licence sheet



On your own, in silence. 5 questions.
You need 4 out of 5 to pass.

3 min

SECTION B

# Your Team and First Job

Four roles, one client, one job.

# Team Roles

Rotate each task. In a group of 3, one pupil gets 2 tasks.



**Safety**
reads the routine, signs the Air Permit, says Stop



**Builder**
builds the circuit on the kit



**Tester**
builds and tests it in the simulation first



**Recorder**
reads out the results for Compare

# Your Client: Harbourlight Adventure Park

## Client

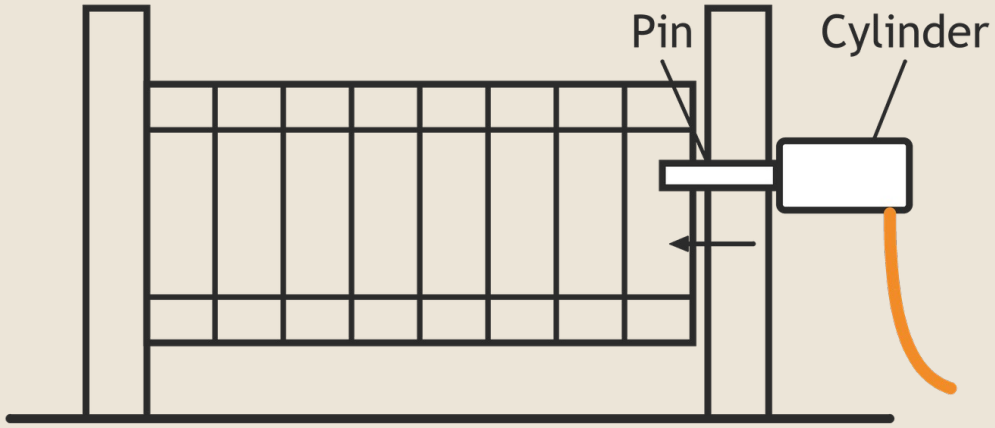
Jo Kerr, Harbourlight's chief engineer

## Job 1

the ticket gate: a button unlocks it

## Done when

the pin goes out, then back in



SECTION C

# The Ticket Gate

Simulate. Air Permit. Then the kit.

# The Gate Circuit

## Regulator

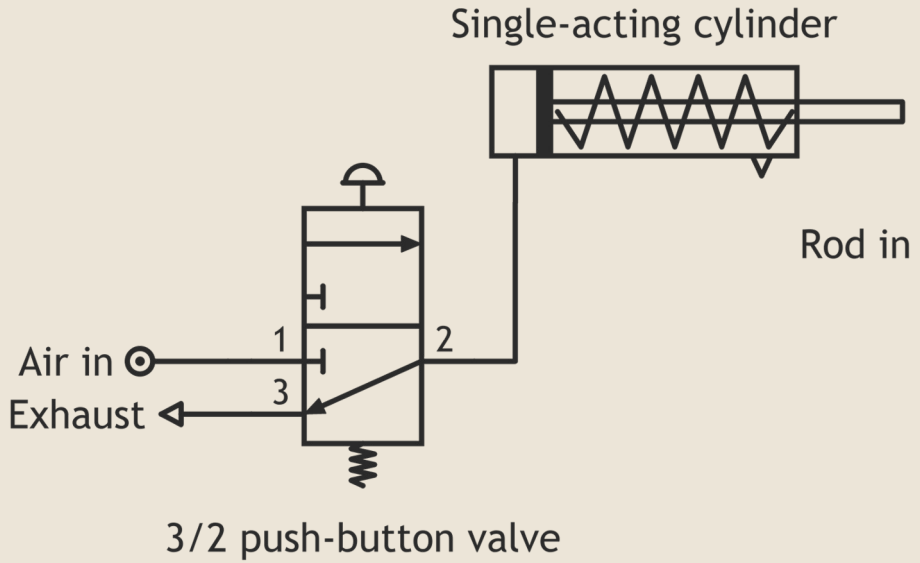
how hard the air pushes

## 3/2 push-button valve

input: 3 ports, 2 states

## Single-acting cylinder

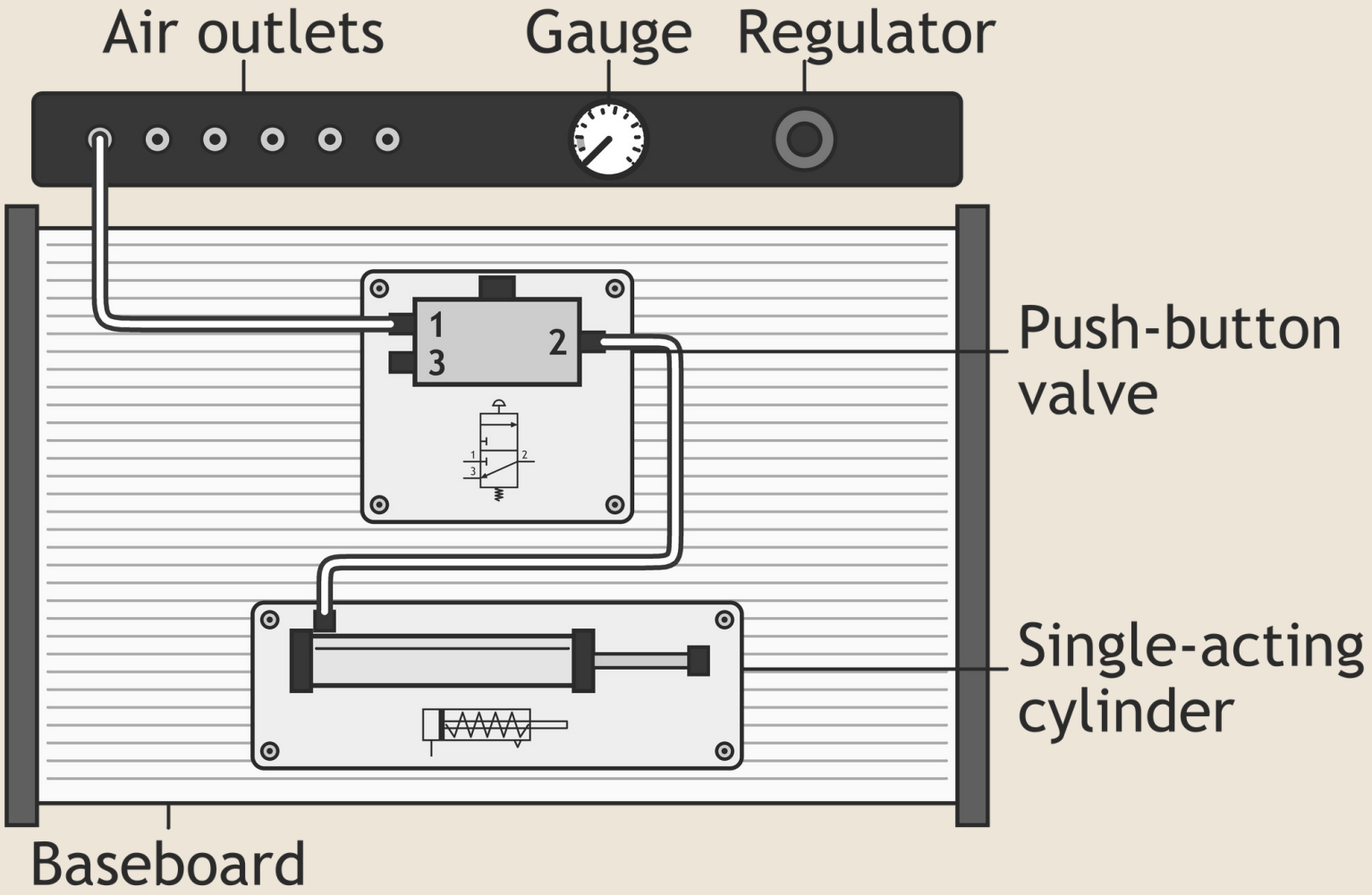
output: air pushes the rod out




3/2 push-button valve

Air supply: the circle with a dot, where the air comes in.

# The Kit, Seen From Above



# Predict

1. You press and hold the button. What does the rod do?

It moves out, and stays out

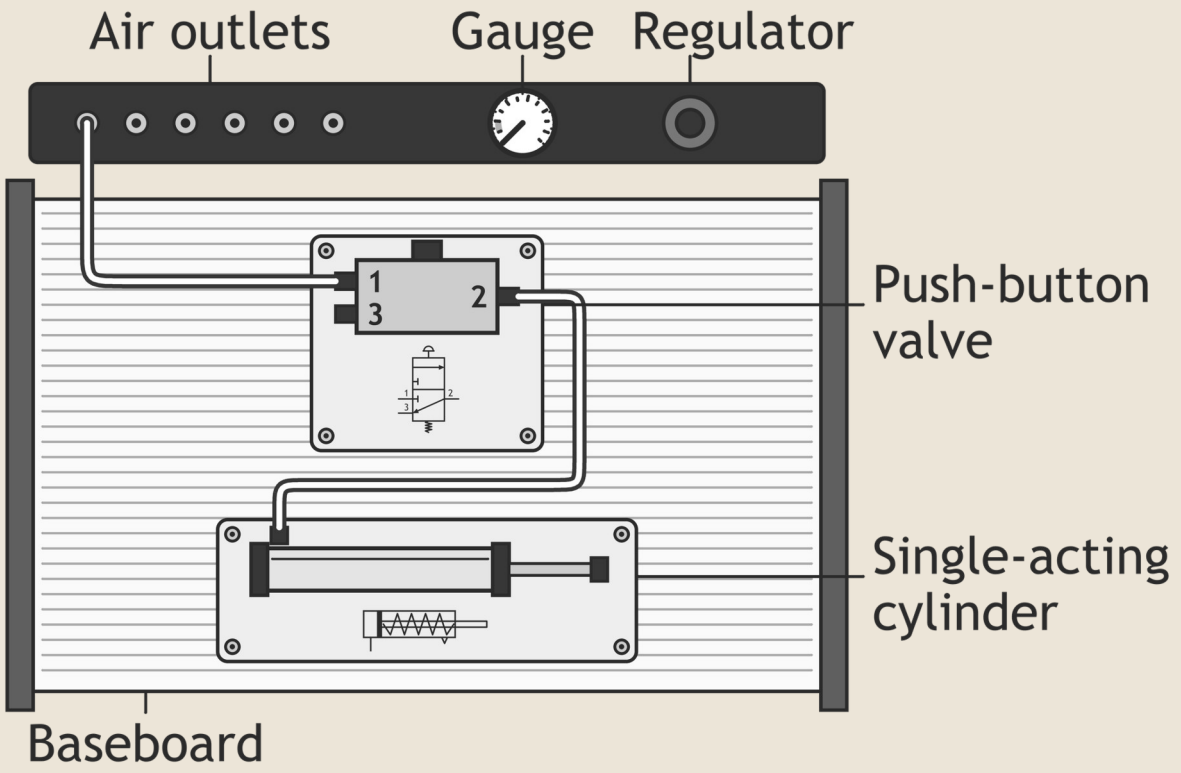
2. You let go of the button. What does the rod do?

The spring pulls it back in

# What to Collect

Builder and Safety: one set per team.

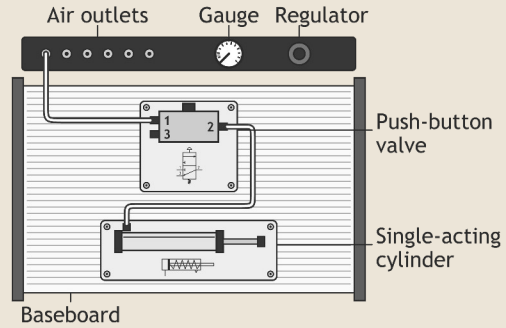
1 Bench card 1
2 Manifold (air supply)
3 3/2 push-button valve
4 Single-acting cylinder
5 Tubing
6 Goggles, one pair each



# Simulate and build

## Tasks 1 and 2, page 2

1 Tester and Recorder: run the simulation.
2 Builder and Safety: collect, then build.
3 Safety: check the safety list.
4 Teacher check, then the air goes on.

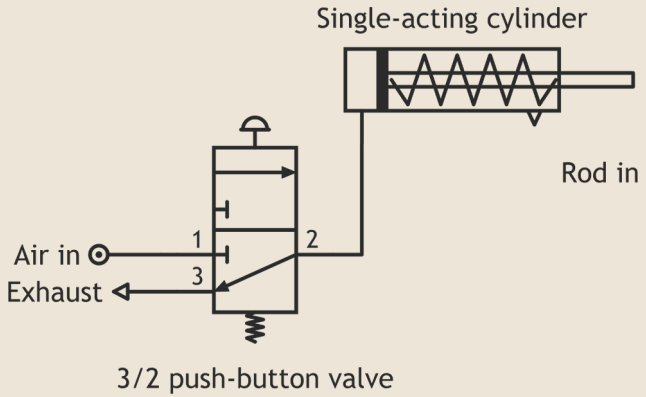



10 min

# Compare and Questions

## Tasks 3 to 5
## booklet page 3

Task 3: fill in Compare.
Task 4: Questions 1 to 3 on your own.
Task 5: Key words.




**5 min**

# Answers: Questions and Key Words

| | | |
|---|---|---|
| 1 | 1 Input and output of the gate circuit | Valve in; cylinder out |
| 2 | 2 Press and hold, then let go | Out and stays out; then back in |
| 3 | 3 Why regulator to 0 before a change? | Nothing moves or blows off |
| 4 | Key words | squashed, regulator, bar, anyone |

# Plenary: Safety First and Last

| | | |
|---|---|---|
| 1 | What pressure do we test at? | Between 1 and 2 bar |
| 2 | What must you do before you change a tube? | Regulator to 0 |
| 3 | Which part is the input in the gate circuit? | The push-button valve |

# Plenary: Safety First and Last

4 What pulls the rod back in?

A spring

5 What must a circuit do before it gets air?

Work in the simulation

# Look back — by the end of this part…



1. I can follow the safety rules and the regulator routine
2. I can say what the rod does when I press and let go
3. I can test a circuit in the simulation before I build it

LOOKING AHEAD

# Lesson 2: Reading the Diagram

Symbols, ports and states: what happens when you let go?

# Credits

**The 5 Safety Rules:** Safety rules and the Lesson 1 circuit adapted from the Matrix CP2080 Pneumatics Essentials worksheets (© Matrix Technology Solutions Limited). Drawings by R Stewart.

R Stewart - October 2026